

SOUP-RUS-DPO: Training, Verification, and Independent Evaluation

Russian DPO fine-tuning of Vikhr-Llama3.1-8B-Instruct-R-21-09-24 using LoRA and Soup CLI

1. Objective and experimental setup

Objective. Evaluate whether DPO fine-tuning produces a valid, updated LoRA adapter and whether the resulting model shows measurable preference improvement beyond the base model.

Base: Vikhrmodels/Vikhr-Llama3.1-8B-Instruct-R-21-09-24 **Method:** DPO + LoRA **Quantization:** 4-bit

Dataset: 761 rows after cleaning. Deterministic shuffle (random_state=42): 500 supplied training examples and 261 independent assessment examples. Soup split the 500 supplied examples into 450 training and 50 validation examples.

Training	Observed run
Epochs / steps	3 epochs / 339 steps
Learning rate	5e-6
Batch size	1
LoRA	r=16, alpha=32; q_proj/v_proj
Runtime	Colab Tesla T4; ~2 h 26 min; return code 0
Adapter	6,815,744 LoRA parameters

2. Training and artifact verification

The training run completed all 339 steps with return code 0. Recorded training loss fell from approximately 0.6928 to 0.0708. The saved artifact is a PEFT/LoRA model with the expected base-model identity and adapter configuration.

Parameter-update verification: checkpoint-200 → checkpoint-339 found 128/128 tensors changed, with 6,815,680 of 6,815,744 elements changed (fraction 0.99999061). Mean absolute delta was 5.51e-05. This independently verifies material adapter updates.

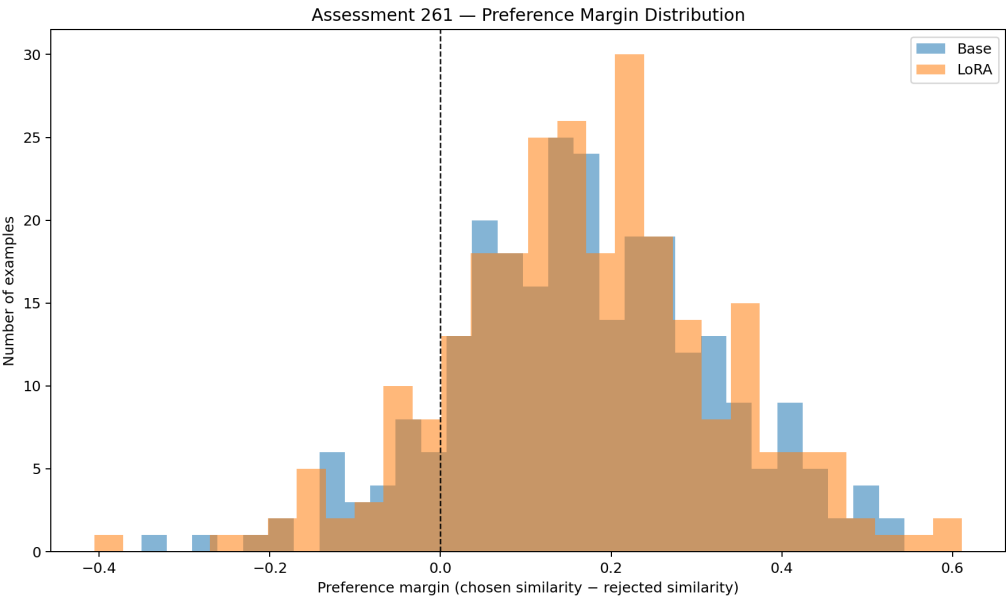


Figure 1. Preference-margin distributions on the 261-example independent assessment. The distributions overlap heavily; the aggregate mean margin changes only slightly.

Interpretation: training and changed adapter weights are verified, but those signals alone do not establish improved generalization.

SOUP-RUS-DPO: Independent Evaluation

The key question is whether the trained adapter improves preference behavior on examples not used for training.

3. Independent 261-example assessment

Metric	Base	LoRA / DPO	Change
Preference accuracy	87.739% (229/261)	87.356% (228/261)	-0.383 pp
Mean preference margin	0.16850	0.16983	+0.00132
Median preference margin	0.16204	0.16739	+0.00536

The transition counts show a near-balanced trade: 10 examples moved from base-wrong to LoRA-correct, while 11 moved from base-correct to LoRA-wrong. The net result is one fewer correct preference on the 261-example assessment.

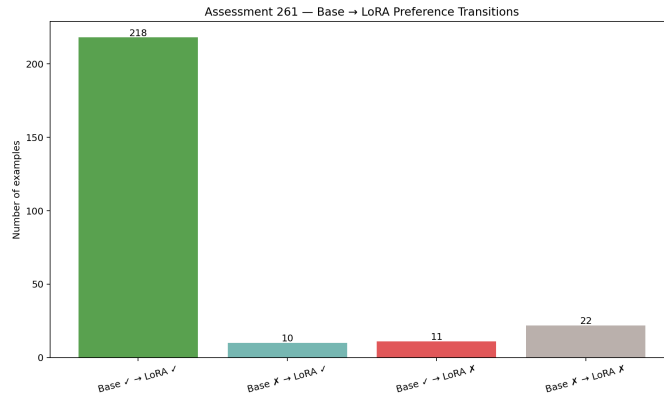


Figure 2. Base-to-LoRA outcome transitions for the 261 independent examples. The largest group remains correct under both models; the two cross-over groups are similar in size.

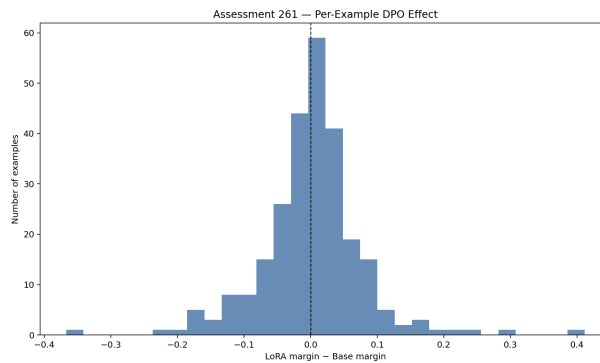


Figure 3. Per-example change in preference margin (LoRA - Base). 137 examples increased and 124 decreased, with no zero-change examples.

4. Validation, control, and limitations

50-example validation subset: base accuracy was 86% (43/50), while LoRA reached 90% (45/50), a +4 percentage-point change. This is a small validation subset and should be read alongside the larger 261-example assessment.

Prompted-base control: on the same 261 assessment, adding the evaluation prompt to the base model produced 89.272% versus 87.739% for the unprompted base (+1.533 pp). Mean margin increased by 0.03117, showing that evaluation prompting itself materially affects the measured result.

Runtime limitation: a direct base-vs-LoRA generation comparison was not valid because the base-model reload encountered a CUDA/CPU device-placement mismatch. LoRA runtime generation passed 5/5. The limitation is recorded rather than treated as a successful comparison.

Conclusion. The run successfully trained and updated a LoRA adapter using DPO, with checkpoint-to-checkpoint parameter changes independently verified. On the 261-example independent assessment, preference accuracy remained essentially flat (87.74% base vs 87.36% LoRA). The experiment therefore validates the training pipeline and artifact integrity, but does not provide evidence of aggregate generalization improvement from this single DPO run.